Improving Search Agent with One Line of Code

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Abstract

Tool-based Agentic Reinforcement Learning (TARL) has emerged as a promising paradigm for training search agents to interact with external tools for multi-turn information-seeking process autonomously . However, we identify a critical training instability that leads to catastrophic model collapse: Importance Sampling Distribution Drift (ISDD). In Group Relative Policy Optimization (GRPO)—a widely adopted TARL algorithm—ISDD manifests as a precipitous decline in the importance sampling ratios, which nullifies gradient updates and triggers irreversible training failure. To address this, we propose **S**earch **A**gent **P**olicy **O**ptimization (**SAPO**), which stabilizes training via a conditional token-level KL constraint. Unlike hard clipping, which ignores distributional divergence, SAPO selectively penalizes the KL divergence between the current and old policies. Crucially, this penalty is applied only to positive tokens with low probabilities where the policy has shifted excessively, thereby preventing distribution drift while preserving gradient flow. Remarkably, SAPO requires only one-line code modification to standard GRPO, ensuring immediate deployability. Extensive experiments across seven QA benchmarks demonstrate that SAPO achieves **+10.6% absolute improvement** (+31.5% relative) over Search-R1, yielding consistent gains across varying model scales (1.5B–14B) and families (Qwen, LLaMA).

1 Introduction

Recent studies Li et al. (2025a); Guo et al. (2024) have introduced search agents capable of interpreting user intent, planning strategies, executing multi-turn actions, and accumulating retrieved evidence for answer generation. Representative methods, such as Search-R1 Jin et al. (2025), employ Group Relative Policy Optimization (GRPO) Kaelbling et al. (1996); Shao et al. (2024) with rule-based rewards to train agents to interact with a search tool over large corpora. Consequently, GRPO has emerged as a prevalent post-training algorithm for Tool-based Agentic Reinforcement Learning (TARL), inspiring numerous follow-up works Mei et al. (2025); Shi et al. (2025a); Qian & Liu (2025); Zhang et al. (2025b).

Despite promising results, existing TARL frameworks face a critical limitation: **standard GRPO introduces Importance Sampling Distribution Drift (ISDD) when the policy shifts outside the trust region.** While several group-based optimization methods (Yu et al., 2025a; Zeng et al., 2025; He et al., 2025; Chen et al., 2025a; An et al., 2025; Wang et al., 2025a; Zhang et al., 2025a) have adopted PPO_clip Schulman et al. (2017) to restrict the deviation between current and old policies. ISDD remains a persistent phenomenon in search agent training, particularly as response lengths or search turns increase. In GRPO, all tokens within a response share an identical advantage value. leading to two primary issues: (1) *Numerical Inaccuracy*: The advantage value may not reflect the true contribution of individual tokens, as different steps contribute unequally to the final answer; and (2) *Misleading Updates*: A response with a correct final answer may contain incorrect intermediate steps (and vice versa). Conflicts arise when the old policy assigns high probability to intermediate correct tokens, but the current policy suppresses them due to negative advantage, leading to vanishing importance sampling weights.

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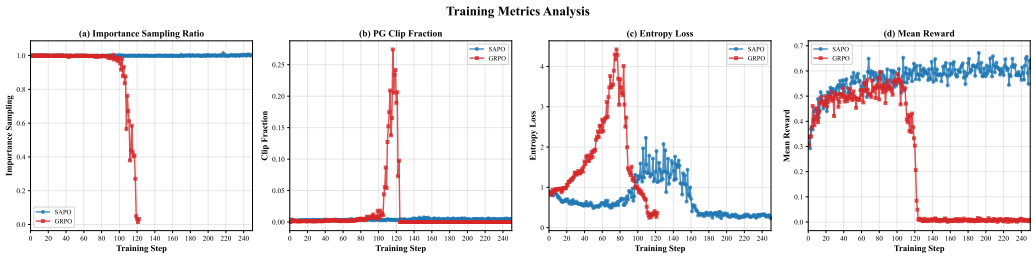

Figure 1: Comparison of training dynamics between **SAPO** and GRPO regarding (a) Importance Sampling Ratio, (b) Clip Ratio, (c) Entropy, and (d) Reward.

Figure.1 depicts the training dynamics of Search-R1 using GRPO, tracking the Importance Sampling (IS) ratios, clip ratio, entropy, and reward. While the IS ratios remain near 1 during the initial phase, they subsequently drop sharply, indicating a growing divergence between the current and old policies. Consequently, the clip ratio exhibits a pronounced spike in the latter stages. Similarly, policy entropy rises substantially at the beginning, suggesting the presence of low-probability positive tokens or high-probability negative tokens. Given the convergence of the IS ratios toward zero, we identify **the suppression of positive tokens (low-probability positive actions)** as the primary driver of ISDD. While the outcome-based reward initially improves, it deteriorates as the IS ratios destabilize and entropy collapses, indicating that late-stage instability compromises final performance. These empirical results demonstrate that despite the use of hard clipping, catastrophic and irreversible model collapse persists.

The primary motivation behind IS and clipping is to correct for distribution shifts and ensure stability. However, for the positive tokens, if the probability under the current policy drops significantly below that of the old policy, the update is effectively suppressed by excessively low importance weights. This yields a weak update signal, preventing the model from recovering valid actions. This observation prompts the following research question:

How can we mitigate optimization conflicts using appropriate constraints when the current policy diverges significantly from the old policy on positive tokens.

To enhance training stability, we propose **S**earch **A**gent **P**olicy **O**ptimization (SAPO), a simple yet theoretically grounded modification of GRPO. Following PPO_KL Schulman et al. (2017), SAPO introduces a conditional KL penalty term to enforce a token-level constraint on the distributional divergence between the current and old policies. To optimize constraint efficiency, SAPO employs an asymmetric mechanism that selectively penalizes conflicting positive tokens and preserves informative gradients from positive tokens. Consequently, SAPO functions as a soft trust region constraint: instead of hard-clipping, we softly penalize large IS ratios, allowing adaptive step sizes while maintaining distributional proximity.

We summarize our contributions as follows.

- We propose SAPO, a policy optimization method designed to stabilize post-training for autonomous multi-turn search agents tackling complex, real-world questions.
- We introduce a conditional KL penalty term that enforces a token-level constraint on the distributional divergence, specifically targeting low-probability positive tokens.
- We demonstrate the effectiveness and generalizability of SAPO across diverse search agents, achieving improvement performance on seven challenging QA benchmarks.

2 Preliminaries

2.1 Proximal Policy Optimization

Proximal Policy Optimization (PPO) Schulman et al. (2017) is one of the most prominent policy gradient methods in reinforcement learning (RL) Kaelbling et al. (1996). PPO facili-

tates stable training by utilizing IS for off-policy updates while constraining the deviation between the current and old policies. The standard unclipped PPO objective is defined as:

$$\mathcal{J}_{\text{PG}}(\theta) = \mathbb{E}_{q\sim\mathcal{D}, o\sim\pi_{\theta_{\text{old}}}(\cdot|q)}\left[r_t(\theta)A_t\right], \quad \text{where} \quad r_t(\theta) = \frac{\pi_\theta(o_t \mid q, o_{<t})}{\pi_{\theta_{\text{old}}}(o_t \mid q, o_{<t})} \tag{1}$$

Here, $\pi_{\theta_{\text{old}}}$ and π_θ denote the old policy and the current policy, respectively, while A_t represents the advantage function. q denotes a query sampled from the data distribution $\mathcal{D}$, and o represents a response generated by the old policy. Direct optimization of this objective can lead to excessive deviation in the IS ratios $r_t(\theta)$, resulting in training instability. To mitigate this, PPO introduces a clipped surrogate objective that imposes a local trust region:

$$\mathcal{J}_{\text{PPO}}(\theta) = \mathbb{E}_{q\sim\mathcal{D}, o\sim\pi_{\theta_{\text{old}}}(\cdot|q)}\left[\min\left(r_t(\theta)A_t,\, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\, A_t\right)\right] \tag{2}$$

Here, ϵ is a hyperparameter defining the clipping bounds. This mechanism limits the incentive for the policy to update when the IS ratio deviates significantly, thereby enhancing stability. However, it zeros out the gradient for clipped tokens, potentially allowing the policy to drift beyond the intended trust region without correction.

2.2 Group Relative Policy Optimization

Group Relative Policy Optimization (GRPO) Guo et al. (2025) eliminates the need for a value function critic by employing group-based advantage estimation. Following Search-R1 Jin et al. (2025), External retrieval tokens are masked during search agent training. Consequently, the GRPO objective is optimized exclusively over the agent-generated tokens:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{q\sim\mathcal{D}, \{o_i\}_{i=1}^{G}\sim\pi_{\theta_{\text{old}}}(\cdot|q)}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\min\left(r_{i,t}(\theta)\hat{A}_{i,t}, \text{clip}\left(r_{i,t}(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_{i,t}\right)\right] \tag{3}$$

where the IS ratio $r_{i,t}(\theta)$ and the advantage $\hat{A}_{i,t}$ are defined as:

$$r_{i,t}(\theta) = \frac{\pi_\theta(o_{i,t} \mid q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})}, \qquad \hat{A}_{i,t} = \hat{A}_i = \frac{R_i - \text{mean}(\{R_j\}_{j=1}^{G})}{\text{std}(\{R_j\}_{j=1}^{G}))} \tag{4}$$

Here, the advantage is computed by normalizing the sequence-level reward R_i against the group statistics. Notably, GRPO assigns the same advantage value to all tokens.

3 Approach

3.1 Search Agent Formulation

Let q denote the user question, a the generated answer, and π_θ the Large Language Model(LLM). We formulate the standard question-answering task as $a = \pi_\theta(\cdot \mid q)$.

RAG systems enhance this process by retrieving knowledge k from an external corpus $\mathcal{D}$ using a retriever $\mathcal{R}(\cdot)$, subsequently generating an answer conditioned on both q and k. We formulate RAG as $a = \pi_\theta(q \mid k), \quad k = \mathcal{R}(q, \mathcal{D})$.

However, this static workflow is often insufficient for complex search tasks that necessitate iterative reasoning and adaptive retrieval. Recent advancements in search agents enable a more dynamic paradigm where retrieval is interleaved with reasoning.

$$a = \pi_\theta\left(q \mid \{z_i, k_i\}_{i=1}^{T}\right), k_i = \mathcal{R}(sq_i \mid \mathcal{D}), sq_i \in z_i \tag{5}$$

Here, the policy output o comprises intermediate reasoning steps z and the final answer a. sq_i denotes a search query generated within the reasoning step z_i to retrieve intermediate knowledge k_i. The sequence $\{z_i, k_i\}$ represents the interleaved trajectory of reasoning and retrieval knowledge. T is the number of search actions.

3.2 Importance Sampling Distribution Drift

In search agents, optimal strategies are inherently sparse, as only a limited subset of queries yields effective results. Consequently, When the policy π_θ rapidly diverges from the exploration path of $\pi_{\theta_{\text{old}}}$, it may assign negligible probabilities to the sampled actions, causing $\pi_\theta(o_t)$ to be orders of magnitude smaller than $\pi_{\theta_{\text{old}}}(o_t)$, driving $r_t(\theta) \to 0$.

Definition 1 (ISDD)**.** Building upon the IS ratios defined in Eq. 4, we define Importance Sampling Distribution Drift (ISDD) as the event where:

$$\mathbb{P}\left(\prod_{t=1}^{T} r_t(\theta) < \epsilon\right) > \phi \tag{6}$$

for small $\epsilon > 0$ and a probability threshold ϕ. This condition indicates that a significant proportion of trajectories suffer from vanishing cumulative importance weights.

Why ISDD Causes Collapse. In GRPO, the policy gradient is weighted by IS ratios:

$$\nabla_\theta \mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{q\sim\mathcal{D},\{o_i\}_{i=1}^{G}\sim\pi_{\theta_{\text{old}}}(\cdot|q)}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\hat{A}_{i,t} r_{i,t}(\theta)\nabla_\theta \log \pi_\theta(o_{i,t}|q, o_{i,<t})\right] \tag{7}$$

When ISDD occurs ($r_t \to 0$), the gradients vanish regardless of the advantage values, freezing the policy update. Consequently, even high-reward trajectories contribute negligible gradients, hindering the model's ability to learn from successful explorations.

Proposition 1 (ISDD Amplification in Interleaved Multi-Step Interactions)**.** *Let the IS ratios r_t follow a log-normal distribution $\log r_t \sim \mathcal{N}(\mu, \sigma^2)$, with the drift parameter $\lambda = \mu + \sigma^2/2$. Due to the low-entropy and bottleneck nature of tool selection, action tokens a_i exhibit significantly higher sensitivity to policy shifts than reasoning tokens z_i, implying $\lambda_k \ll \lambda_a < 0$.*

The expected cumulative importance weight for the trajectory decomposes as follows:

$$\mathbb{E}\left[\prod_{t=1}^{L} r_t\right] = \mathbb{E}\left[\prod_{t\in\mathcal{Z}} r_t\right] \cdot \mathbb{E}\left[\prod_{t\in\mathcal{A}} r_t\right] = \underbrace{\exp\left(L_z\lambda_z\right)}_{\textit{Reasoning Drift}} \cdot \underbrace{\exp\left(L_a\lambda_a\right)}_{\textit{Interaction Drift}} \tag{8}$$

*where L denote the total number of tokens. ISDD is more severe in agent tasks than in QA tasks because the **Interaction Drift** term decays exponentially with the volume of actions L_a, driven by the highly negative drift parameter λ_a.*

3.3 Search Agent Policy Optimization

To alleviate ISDD and enhance training stability, SAPO introduces an auxiliary penalty term. This term enforces a constraint on the distributional divergence between the current and old policies. The SAPO token-level objective is formulated as:

$$\begin{aligned}\mathcal{J}_{\text{SAPO}}(\theta) =& \mathbb{E}_{q\sim\mathcal{D},\{o_i\}_{i=1}^{G}\sim\pi_{\theta_{\text{old}}}(\cdot|q)}\Bigg[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|} \\ & \min\left(r_{i,t}(\theta)\hat{A}_{i,t}, \text{clip}\left(r_{i,t}(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_{i,t}\right) \underbrace{+\gamma \text{KL}\left[\pi_\theta \parallel \pi_{\text{old}}\right]}_{\text{Drift Penalty}}\Bigg]\end{aligned} \tag{9}$$

where γ denotes the penalty coefficient. Unlike the KL penalty in standard RL, which typically regulates divergence against a fixed reference model, SAPO penalizes deviations from the dynamic old policy to prevent the vanishing gradient problem. We approximate the KL divergence using the log-ratio:

$$\mathrm{KL}\left[\pi_\theta \,\|\, \pi_{\text{old}}\right] = \log \frac{\pi_\theta(o_{i,t} \mid q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})} = \log r_{i,t}(\theta) \tag{10}$$

Standard KL penalties may indiscriminately hinder policy exploration. To address this, we propose a conditional KL penalty that activates only when the probability ratio drops significantly for samples with positive advantages. The conditional term is defined as:

$$\mathrm{KL}_{cond}\left[\pi_\theta \,\|\, \pi_{\text{old}}\right] = \mathbb{I}(r_{i,t}(\theta) < \tau, \hat{A}_{i,t} > 0) \log r_{i,t}(\theta) \tag{11}$$

Here, $\mathbb{I}(\cdot)$ is the indicator function and τ is the threshold to identify drifting tokens. The penalty term has three crucial properties:

1. **Conditional**: Only applied when $A_t > 0$ (positive advantage)
2. **Threshold-gated**: Only triggered when $r_t < \tau$ (excessive shift)
3. **Logarithmic**: $\log r_t$ grows slowly, allowing gradual exploration

3.4 Gradient Analysis

We derive the gradient of the SAPO auxiliary term as follows:

$$\begin{aligned}
\nabla_\theta \mathcal{J}_{\text{SAPO_KL}}(\theta) &= \nabla_\theta \mathbb{I}(r_{i,t}(\theta) < \tau, \hat{A}_{i,t} > 0) \log r_{i,t}(\theta) \\
&= \mathbb{I}(\hat{A}_{i,t} > 0) \nabla_\theta \left[\mathbb{I}(r_{i,t}(\theta) < \tau) \log r_{i,t}(\theta)\right] \\
&= \mathbb{I}(\hat{A}_{i,t} > 0)[\mathbb{I}(r_{i,t}(\theta) < \tau) \nabla_\theta \log r_{i,t}(\theta) + \nabla_\theta \mathbb{I}(r_{i,t}(\theta) < \tau) \log r_{i,t}(\theta)] \\
&= [\mathbb{I}(r_{i,t}(\theta) < \tau, \hat{A}_{i,t} > 0) - \delta(r_{i,t}(\theta) - \tau) r_{i,t}(\theta) \log r_{i,t}(\theta)] \nabla_\theta \log \pi_\theta(o_{i,t}|q, o_{i,<t})
\end{aligned} \tag{12}$$

Consequently, the total gradient of the SAPO objective is:

$$\begin{aligned}
\nabla_\theta \mathcal{J}_{\text{SAPO}}(\theta) = \mathbb{E}_{q \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \Bigg[\frac{1}{G} \sum_{i=1}^{G} \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} (\hat{A}_{i,t} r_{i,t}(\theta) \\
\underbrace{+\gamma[\mathbb{I}(r_{i,t}(\theta) < \tau, \hat{A}_{i,t} > 0) - \delta(r_{i,t}(\theta) - \tau) r_{i,t}(\theta) \log r_{i,t}(\theta)]}_{\text{Drift Penalty Gradient Coefficient}}) \nabla_\theta \log \pi_\theta(o_{i,t}|q, o_{i,<t}) \Bigg].
\end{aligned} \tag{13}$$

where $\delta(\cdot)$ denotes the Dirac delta function, arising from the differentiation of $\mathbb{I}(\cdot)$.

4 Experiments

4.1 Experiment Settings

Datasets. We evaluate our method on seven diverse QA benchmarks, categorized into single-hop or multi-hop retrieval tasks. The single-hop datasets include Natural Questions (NQ) Kwiatkowski et al. (2019), TriviaQA Joshi et al. (2017), and PopQA Mallen et al. (2022). The multi-hop datasets comprise HotpotQA Yang et al. (2018), 2WikiMultihopQA Ho et al. (2020), Musique Trivedi et al. (2022b), and Bamboogle Press et al. (2022). We employ Exact Match (EM) accuracy as the evaluation metric across all datasets. Following Search-R1 Jin et al. (2025), SAPO is trained on acomposite dataset consisting of NQ and HotpotQA.

Table 1: Accuracy comparison of SAPO-3B against baseline methods using Qwen2.5-3B Qwen et al. (2025) across multiple QA benchmarks. **Bold** indicates best results, and underline denotes second best results.

Methods	Single-Hop QA				Multi-Hop QA					Avg.
	NQ	TriviaQA	PopQA	*Avg.*	HotpotQA	2Wiki	Musique	Bamboogle	*Avg.*	
w/o Retrieval										
Direct Generation	0.106	0.288	0.108	0.167	0.149	0.244	0.020	0.024	0.109	0.134
SFT	0.249	0.292	0.104	0.215	0.186	0.248	0.044	0.112	0.148	0.176
R1-Instruct Guo et al. (2025)	0.210	0.449	0.171	0.277	0.208	0.275	0.060	0.192	0.184	0.224
R1-Base Guo et al. (2025)	0.226	0.455	0.173	0.285	0.201	0.268	0.055	0.224	0.187	0.229
Workflow w/ Retrieval										
Naive RAG Lewis et al. (2020)	0.348	0.544	0.387	0.426	0.255	0.226	0.047	0.080	0.152	0.270
IRCoT Trivedi et al. (2022a)	0.111	0.312	0.200	0.208	0.164	0.171	0.067	0.240	0.161	0.181
Agent w/ Retrieval										
Search-o1 Li et al. (2025c)	0.238	0.472	0.262	0.324	0.221	0.218	0.054	0.320	0.203	0.255
Search-R1-Instruct Jin et al. (2025)	0.397	0.565	0.391	0.451	0.331	0.310	0.124	0.232	0.249	0.336
Search-R1-Base Jin et al. (2025)	0.421	0.583	0.413	0.472	0.297	0.274	0.066	0.128	0.191	0.312
ReSearch-Instruct Chen et al. (2025b)	0.365	0.571	0.395	0.444	0.351	0.272	0.095	0.266	0.246	0.331
ReSearch-Base Chen et al. (2025b)	0.427	0.597	0.430	0.485	0.305	0.272	0.074	0.128	0.195	0.319
ZeroSearch-Base Sun et al. (2025)	0.430	0.616	0.414	0.487	0.338	0.346	0.130	0.139	0.238	0.345
StepSearch-Base Wang et al. (2025b)	–	–	–	–	0.329	0.339	0.181	0.328	0.294	–
EXSEARCH-Base Shi et al. (2025b)	0.368	–	–	–	0.422	0.372	0.138	–	–	–
O^2-Searcher Mei et al. (2025)	0.444	0.597	0.429	0.490	0.388	0.374	0.160	0.344	0.317	0.391
AutoRefine-Instruct Shi et al. (2025a)	0.436	0.597	0.447	0.493	0.404	0.380	0.169	0.336	0.322	0.396
AutoRefine-Base Shi et al. (2025a)	0.467	0.620	0.450	0.512	0.405	0.393	0.157	0.344	0.325	0.405
InForage Qian & Liu (2025)	0.421	0.597	0.452	0.490	0.409	0.428	0.172	0.360	0.342	0.405
CriticSearch Zhang et al. (2025b)	–	–	–	–	0.414	0.409	0.180	0.368	0.343	–
SE-Search Li et al. (2026)	**0.475**	0.624	0.423	0.507	0.450	0.361	0.183	0.424	0.355	0.420
SAPO-3B-Base (Ours)	0.474	**0.630**	0.449	0.517	0.449	0.412	0.196	0.424	0.370	0.433
SAPO-3B-Instruct (Ours)	0.469	0.629	**0.457**	**0.518**	**0.456**	**0.448**	**0.203**	**0.432**	**0.396**	**0.442**

Baselines. We compare SAPO against three categories of methods: (1) **Retrieval-free methods**: direct LLM generation, Supervised Fine-Tuning (SFT), and R1-style training Guo et al. (2025) without retrieval; (2) **Retrieval-enhanced workflows**: Naive RAG, which retrieves documents based solely on the input question, and IRCoT Trivedi et al. (2022a), which interleaves retrieval with Chain-of-Thought; (3) **Retrieval-augmented agents**: recent agentic methods including Search-o1 Li et al. (2025c), Search-R1 Jin et al. (2025), ReSearch Chen et al. (2025b), ZeroSearch Sun et al. (2025), StepSearch Wang et al. (2025b), EXSEARCHShi et al. (2025b), O^2-Searcher Mei et al. (2025), InForage Qian & Liu (2025), CriticSearch Zhang et al. (2025b), AutoRefine Shi et al. (2025a) and SE-Search Li et al. (2026).

Implementation Details. We utilize the external corpus Karpukhin et al. (2020) used by Search-R1, with E5-base-v2 Wang et al. (2022) serving as the retrieval engine. Unless otherwise specified, the retriever returns the top-3 documents per query and the maximum number of search turns is set to $T_{max} = 5$. We employ the Qwen2.5 series Qwen et al. (2025) as the backbone language model. In the RL phase, we employ a simple rule-based reward function. Specifically, this is an outcome-oriented reward calculated using the F1-score between the predicted and ground-truth answers. the KL penalty coefficient is set to $\gamma = 0.1$. The IS ratios threshold is set to $\tau = 1.0$.

4.2 Main Performance

Table 1 presents the main experimental results comparing SAPO against baseline methods. The Avg. column reports the average accuracy. All methods utilize the same retriever, knowledge corpus, training data, and LLM backbone (Qwen2.5-3B).

SAPO outperforms other methods. SAPO surpasses Search-R1 and state-of-the-art methods (*i.e.*, AutoRefine and CriticSearch) across all seven benchmarks. Our method achieves an average EM accuracy of 0.442, demonstrating that the proposed SAPO post-training strategy substantially enhances the search agent's capabilities. Specifically, SAPO yields a significant absolute improvement of 10.6 points (a relative gain of 31.5%) over the Search-R1 baseline.

Method	Single-Hop QA			Multi-Hop QA				Avg.
	NQ	TriviaQA	PopQA	HotpotQA	2Wiki	Musique	Bamboogle	
Search-R1-GRPO	0.402	0.662	0.436	0.374	0.376	0.134	0.333	0.388
+ KL	0.425	0.625	0.443	0.428	0.371	0.175	0.342	0.398
	(+0.023)	(-0.037)	(+0.007)	(+0.054)	(-0.005)	(+0.041)	(+0.009)	(+0.010)
+ KL_r	0.439	0.660	0.456	**0.457**	0.369	0.169	0.369	0.417
	(+0.014)	(+0.035)	(+0.013)	(+0.029)	(-0.002)	(-0.006)	(+0.027)	(+0.019)
+ KL_{ra} (SAPO)	**0.442**	**0.673**	**0.474**	0.433	**0.415**	**0.181**	**0.382**	**0.429**
	(+0.003)	(+0.013)	(+0.018)	(-0.024)	(+0.046)	(+0.012)	(+0.013)	(+0.012)

Table 2: Ablation results on different KL terms in **SAPO**. The values in parentheses denote the absolute performance change compared to the previous row.

SAPO demonstrates particularly strong gains on multi-hop QA benchmarks. Performance improvements are most pronounced on complex, multi-hop QA tasks. For instance, compared to the CriticSearch, SAPO improves performance on HotpotQA by 4.2 percentage points (a 10.1% relative improvement) and on Bamboogle by 6.4 percentage points (a 17.4% relative improvement). Furthermore, across all multi-hop benchmarks, SAPO achieves an average increase of 5.3 percentage points (a 15.45% relative improvement). compared to Search-R1, SAPO achieves a substantial increase of 14.7 percentage points (a 24.9% relative improvement) across the four multi-hop QA benchmarks. These gains are attributed to the training stability provided by our proposed auxiliary KL penalty term.

4.3 Ablation Studies

We conduct ablation study to evaluate the impact of SAPO's key components. To ensure a fair comparison, all models are trained for the same 300 steps, sampling 5 rollout responses. The baseline, Search-R1, utilizes Qwen2.5-3B-Instruct as the language model and the F1 score as the outcome-based reward. Table 2 details the accuracy across seven benchmarks and quantifies the contribution of each component. We compare four configurations: (1)**GRPO**: The baseline GRPO algorithm as employed in Search-R1. (2)**GRPO w/ KL**: GRPO augmented with an unconditional KL penalty term. (3)**GRPO w/ KL_r**: GRPO with a KL penalty conditioned solely on the IS ratio. (4)**GRPO w/ KL_ra (SAPO)**: The complete SAPO configuration, where the KL penalty is conditioned on both the IS ratios and the advantage.

Impact of Selective KL Penalties. Table.2 details the performance progression, starting with the Search-R1-GRPO baseline, which achieves an average accuracy of 0.388. Incorporating a simple unconditional KL term yields a significant performance on Musique (+30.6%) and HotpotQA (+14.4%). This results in a modest increase in average accuracy to 0.398. The ratio-conditioned variant (KL_r) further improves stability, raising the average to 0.417 and notably improving the HotpotQA score to 0.457. Finally, the full SAPO configuration (KL_ra) achieves the best overall balance, securing the highest scores on six out of seven benchmarks with an average of 0.429, an absolute increase of 0.041 over the baseline (about 10.6% relative). These findings validate that the conditional KL penalty in SAPO effectively balances exploration and constraint, leading to consistent aggregate improvements.

4.4 Detailed Analysis

Hyperparameter Sensitivity Analysis. We analyze the sensitivity of the IS threshold t within our SAPO method. The validation EM scores in Figure 2(a) indicate that $t = 1.0$ consistently yields the best performance throughout the training process. Specifically, at step 280, the model achieves an EM score of 0.442. In contrast, $t = 1.2$ results in a final score of 0.434, while $t = 0.6$ and $t = 0.8$ lead to inferior results of 0.418 and 0.425, respectively. Notably, the performance divergence among different threshold values widens as training progresses, suggesting that the selection of t becomes increasingly critical in later stages. These results confirm that $t = 1.0$ strikes an optimal balance between exploration and exploitation, thereby being adopted as the default setting for all subsequent experiments.

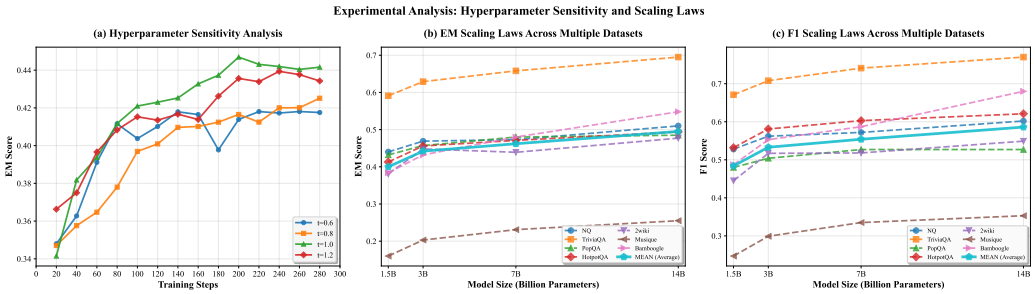

Figure 2: (a) Hyperparameter sensitivity analysis. (b,c) Scaling trends of Qwen2.5-Instruct with **SAPO** across different model sizes (1.5B to 14B). We report both EM and F1 scores.

Method	Single-Hop QA			Multi-Hop QA				Avg.
	NQ	TriviaQA	PopQA	HotpotQA	2Wiki	Musique	Bamboogle	
Backbone: LLaMA-3.2-3B-Base								
Search-R1-GRPO	0.394	0.596	0.437	0.280	0.264	0.056	0.105	0.305
Search-R1-SAPO	**0.471**	**0.630**	**0.452**	**0.355**	**0.301**	**0.102**	**0.248**	**0.366**
Backbone: LLaMA-3.2-3B-Instruct								
Search-R1-GRPO	0.357	0.578	0.378	0.314	0.233	0.090	0.306	0.322
Search-R1-SAPO	**0.475**	**0.634**	**0.446**	**0.444**	**0.386**	**0.187**	**0.384**	**0.422**

Table 3: Generalization results on LLaMA-3.2-3B (Base and Instruct). **SAPO** consistently outperforms the GRPO baseline across both model variants.

Scaling with Model Size. We investigate the scalability of SAPO by evaluating its performance across backbone models of varying sizes, specifically the Qwen2.5-Instruct series ranging from 1.5B to 14B parameters. Figure. 2 reports the EM and F1 scores. We observe a strong positive correlation between model size and performance, aligning with established scaling laws Kaplan et al. (2020). As the parameter count increases, SAPO demonstrates a monotonic improvement in average accuracy. Notably, scaling from 1.5B to 14B results in a substantial performance leap, raising the average EM from 0.400 to 0.495 and the F1 score from 0.484 to 0.586. This confirms that SAPO effectively leverages the enhanced reasoning capabilities inherent in LLM.

Robustness across Model Families. To assess the generalizability of SAPO across different architectures, we extend our evaluation to the LLaMA-3.2Touvron et al. (2023) series, testing both the Base and Instruct versions of the 3B model. As detailed in Table 3, SAPO consistently outperforms the GRPO baseline across both variants. On LLaMA-3.2-3B-Base, SAPO improves the average accuracy from 0.305 to 0.366, proving its capacity to align base pretrained models effectively. The gains are even more pronounced on LLaMA-3.2-3B-Instruct, where SAPO boosts average accuracy by 10.0 percentage points (from 0.322 to 0.422). These results indicate that SAPO is a model-agnostic framework capable of enhancing search capabilities regardless of the underlying model family or its prior alignment stage.

5 Conclusion

Tool-based agentic reinforcement learning, such as GRPO, introduces Importance Sampling Distribution Drift (ISDD). This drift often causes the importance sampling ratio to decline sharply, leading to training instabilities that manifest as catastrophic and irreversible model collapse. In this paper, we propose SAPO (**S**earch **A**gent **P**olicy **O**ptimization) to enhance the training stability of search agents. We analyze the phenomena of increasing clipping rates and decreasing rewards, attributing these effects to ISDD. To mitigate this, SAPO introduces a penalty term that enforces a token-level conditonal constraint on the distributional divergence between the current and old policies. Extensive experiments on single-hop

and multi-hop QA benchmarks demonstrate that SAPO significantly outperforms existing methods, improving absolute performance by 10.6 points over the Search-R1 baseline.

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A Appendix

A.1 Related Works

Retrieval-Augmented Generation Large Language Models (LLMs) Zhao et al. (2023) have demonstrated remarkable capabilities in natural language understanding, reasoning, and information synthesis. However, they lack inherent access to real-time external knowledge and remain susceptible to hallucinations and factual errors. Retrieval-Augmented Generation (RAG) Lewis et al. (2020) addresses these limitations by integrating LLMs with retrieval engines, conditioning generation on retrieved external passages to mitigate inaccuracies. Despite their utility, static RAG pipelines typically employ a "retrieve-then-generate" paradigm, which restricts the LLM's autonomy in determining *when* and *what* to search. This rigidity limits their adaptability in complex, real-world applications where iterative information seeking is required.

Search Agents While traditional search engines (e.g., Google, Bing) facilitate efficient access to ranked web pages, they often struggle to capture nuanced user intent in complex queries Kobayashi & Takeda (2000). Search agents Li et al. (2025a; 2026) represent an evolution beyond static RAG by invoking search tools within a multi-step reasoning process, allowing for autonomous query refinement and iterative retrieval. Systems such as Search-o1 Li et al. (2025c), WebThinker Li et al. (2025d), and WebDancer Wu et al. (2025) exemplify this capability. Concurrently, research has focused on training strategies to optimize search behaviors, including Search-R1 Jin et al. (2025), ReSearch Chen et al. (2025b), AutoRefine Shi et al. (2025a), SE-Search Li et al. (2026) and WebSailor Li et al. (2025b). However, a pervasive limitation among these methods is post-training instability, which frequently manifests as catastrophic model collapse or irreversible performance degradation. To mitigate this, we propose Search Agent Policy Optimization (SAPO), a method designed to enhance training stability through constrained policy updates.

Tool-based Agentic Reinforcement Learning Reinforcement Learning (RL) provides a robust framework for aligning LLM behaviors with human intent. Prominent approaches include Proximal Policy Optimization (PPO) Schulman et al. (2017), Direct Preference Optimization (DPO) Rafailov et al. (2023), and Group Relative Policy Optimization (GRPO) Shao et al. (2024). GRPO is particularly notable for its use of group-based normalization, which eliminates the need for a separate value network. Extensions such as Dynamic Sampling Policy Optimization (DAPO) Yu et al. (2025b) and Group Sequence Policy Optimization (GSPO) Zheng et al. (2025) have further refined this paradigm. Other group-based optimization methods include those detailed in Zeng et al. (2025); He et al. (2025); Chen et al. (2025a); An et al. (2025); Wang et al. (2025a); Zhang et al. (2025a). Nevertheless, applying RL to tool-augmented agentic environments remains highly unstable due to the interactive, multi-turn nature of the tasks Wang et al. (2026); Deng et al. (2025a;b). Addressing this challenge, our work introduces a sample-based penalty term to enforce a token-level constraint between the updated and reference policies, thereby stabilizing the training of search agents.

A.2 Training and Evaluation Datasets

Consistent with Search-R1 and other RL-based search agents, we construct our training dataset by aggregating NQ Kwiatkowski et al. (2019) and HotpotQA Yang et al. (2018), yielding a total of 169,615 samples. For evaluation, we construct a comprehensive benchmark of 51,713 samples, incorporating the test sets of four datasets (NQ Kwiatkowski et al. (2019), TriviaQA Joshi et al. (2017), PopQA Mallen et al. (2022), Bamboogle Press et al. (2022)) and the development sets of three multi-hop datasets (HotpotQA Yang et al. (2018), 2WikiMultihopQA Ho et al. (2020), and Musique Trivedi et al. (2022b)).

A.3 Search Agent Prompt

Figure 3 illustrates the prompt template used for the SAPO search agent. The template instructs the assistant to perform multi-turn search-driven reasoning by alternating be-

You are a helpful assistant excel at answering questions with multi-turn search engine calling. To answer questions, you must first reason through the available information using <think> and </think>. If you identify missing knowledge, you may issue a search request using <search> query </search> at any time. The retrieval system will provide you with the three most relevant documents enclosed in <documents> and </documents>. You may send multiple search requests if needed. Once you have sufficient information, provide a concise final answer using <answer> and </answer>. For example, <answer> Donald Trump </answer>. Question: {**question**}

Figure 3: Prompt template for SAPO.

tween internal thought processes in <think> </think> and query generation in <search> </search>. Retrieved results are returned as up to three documents within <documents> </documents>. The design permits multiple search calls and directs the agent to consolidate evidence into a concise final response enclosed by <answer> </answer>.

A.4 SAPO Hyperparameters

We implement our method using the VeRL framework Sheng et al. (2025). As detailed in Table 4, we maintain consistency with Search-R1 and SE-Search regarding data, actor, and rollout configurations.Specifically for SAPO, the KL penalty coefficient is set to $\gamma = 0.1$, and the IS ratios threshold is set to $\tau = 1.0$.

Table 4: Hyperparameters used in SAPO.

Module	Hyper-parameter	Value
Data	Max documents length	512
	Max response length	2048
	Total training steps	420
	Retriever top-k	3
Actor	Training batch size	256
	Micro batch size	64
	Learning rate	1×10^{-6}
	KL coefficient β	0.001
	Clip ratio ϵ	0.2
Rollout	Max search actions	5
	Group size G	10
	Temperature	1.0
	Top-p	0.95
KL term	KL penalty coeff γ	0.1
	IS ratios threshold τ	1.0

A.5 Generalization and Scalability

We investigate the scalability of SAPO by evaluating its performance across backbone models of varying capacities, specifically the Qwen2.5-Instruct series ranging from 1.5B to 14B parameters. To isolate the impact of model scale, we keep the retriever, knowledge corpus, and evaluation protocols constant. Table 5 reports the Exact Match (EM) and F1 scores. The results exhibit a clear positive correlation between model size and performance, aligning with established scaling laws Kaplan et al. (2020).

A.6 Implementation Simplicity

We augment the standard GRPO surrogate objective with a concise penalty term to enforce token-level constraints on distributional shifts between the current and old policies. As shown in Figure A.6, the core contribution of SAPO is encapsulated in the `kl_term_loss`

Size	Metric	Single-Hop QA			Multi-Hop QA				Avg.
		NQ	TriviaQA	PopQA	HotpotQA	2Wiki	Musique	Bamboogle	
1.5B	EM	0.440	0.591	0.432	0.413	0.381	0.160	0.384	0.400
	F1	0.528	0.671	0.480	0.532	0.445	0.247	0.487	0.484
3B	EM	0.469	0.629	0.457	0.456	0.448	0.203	0.432	0.442
	F1	0.562	0.708	0.504	0.581	0.517	0.299	0.553	0.533
7B	EM	0.473	0.658	0.480	0.471	0.439	0.231	0.480	0.462
	F1	0.572	0.741	0.527	0.603	0.518	0.335	0.587	0.554
14B	EM	**0.510**	**0.695**	**0.485**	**0.494**	**0.477**	**0.255**	**0.548**	**0.495**
	F1	**0.602**	**0.770**	**0.527**	**0.621**	**0.549**	**0.353**	**0.680**	**0.586**

Table 5: Scaling trends of Qwen2.5-Instruct with **SAPO** across different model sizes (1.5B to 14B). We report both EM and F1 scores. The best results are marked in bold.

expression. This term yields a scalar, token-level divergence penalty that is added to the global loss to directly limit per-token policy drift. By operating at the token granularity and requiring only a single line of code, this modification provides a lightweight and interpretable regularizer that complements the existing PPO clipping mechanism while enabling finer control over distributional changes during policy updates.

```
def compute_policy_loss(old_log_prob, log_prob, advantages, eos_mask,
    cliprange):
   negative_approx_kl = log_prob - old_log_prob
   ratio = torch.exp(negative_approx_kl)
   ppo_kl = verl_F.masked_mean(-negative_approx_kl, eos_mask)

   pg_losses = -advantages * ratio
   pg_losses2 = -advantages * torch.clamp(ratio, 1.0 - cliprange, 1.0 +
       cliprange)

   kl_term_loss = -verl_F.masked_mean(
       torch.log(ratio).masked_fill(ratio > t, 0)[advantages[:, 0] >= 0],
       eos_mask[advantages[:, 0] >= 0]
   )

   pg_loss = verl_F.masked_mean(torch.max(pg_losses, pg_losses2), eos_mask)
   pg_clipfrac = verl_F.masked_mean(torch.gt(pg_losses2, pg_losses).float(),
       eos_mask)
   return pg_loss + w * kl_term_loss, pg_clipfrac, ppo_kl
```

A.7 Additional Experiments Results

To provide a comprehensive characterization of SAPO's training stability, Figure 4 illustrates the SAPO EM evolution of four backbones(Qwen2.5-7B-Instruct; Llama-3.2-3B-Instruct; Qwen2.5-1.5B-Instruct; Qwen2.5-3B-Instruct) across seven benchmarks: Natural Questions (NQ), TriviaQA, PopQA, Bamboogle, HotpotQA, 2Wiki, and Musique.

A.8 Case Studies

Table 6 and 7 present full trajectories generated by the 3B and 14B models using SAPO, based on two randomly selected examples from 2WikiMultihopQA and Musique. These qualitative results demonstrate the agent's capacity for autonomous reasoning and iterative query refinement.

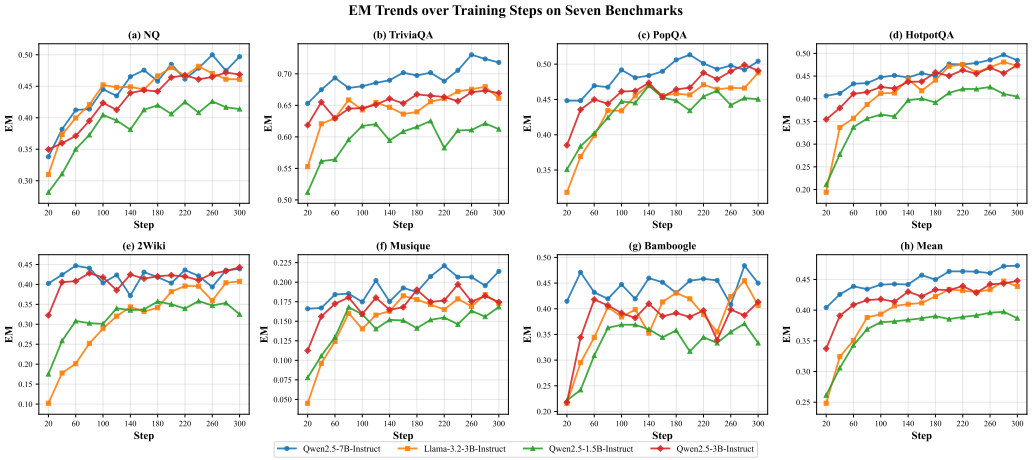

Figure 4: Evolution of EM accuracy over training steps for SAPO across seven benchmarks.

Table 6: Case Studies illustrating SAPO trajectories from the 2wikimultihopqa benchmark. Key evidence is extracted from the original searched documents.

Question: Which film has the director who died earlier, Deuces Wild or Cavalcade Of The West? **Ground Truth:** [”Cavalcade of the West”, ”Cavalcade Of The West”]
Model Output: Qwen2.5-3B-Instruct-SAPO
To determine which film has the director who died earlier between ”Deuces Wild” and ”Cavalcade Of The West”, I need to gather information about the directors of these films and their death dates. Let’s start by searching for this information. <search> **Director of Deuces Wild** </search> <documents> **Doc 1 (Title: ”Deuces Wild”):** Deuces Wild is a 2002 American crime drama film directed by **Scott Kalvert**... **Doc 3 (Title: ”Scott Kalvert”):** Scott Kalvert (August 15, 1964 – **March 5, 2014**) was an American film director, known mainly for his 1995 film ”The Basketball Diaries”... and 2002’s ”Deuces Wild”. </documents> <search> **Director of Cavalcade Of The West** </search> <documents> **Doc 1 (Title: ”Cavalcade of the West”):** Cavalcade of the West is a 1936 American Western film directed by **Harry L. Fraser**... **Doc 3 (Title: ”Cavalcade of the West”):** ...Cavalcade of the West is a 1936 American Western film directed by Harry L. Fraser. </documents> <search> **Harry L. Fraser** </search> <documents> **Doc 1 (Title: ”Harry W. Fraser”):** Harry W. Fraser (7 June 1884 – **13 May 1950**) was an American labor leader... **Doc 3 (Title: ”Harry W. Fraser”):** Fraser was president of the Order of Railway Conductors of America from 1941 to 1950. ... Fraser was born on June 7, 1884 and **died on May 13, 1950**. </documents> <search> **Scott Kalvert** </search> <documents> **Doc 1 (Title: ”Scott Kalvert”):** Scott Kalvert (August 15, 1964 – **March 5, 2014**) was an American film director... **Doc 2 (Title: ”Scott Kalvert”):** ...on March 5, 2014 from an apparent suicide. </documents> Based on the information provided, Scott Kalvert died on March 5, 2014, while Harry L. Fraser died on May 13, 1950. Therefore, the director who died earlier is Harry L. Fraser, and the film ”Cavalcade Of The West” was directed by him. <answer> **Cavalcade Of The West** </answer>

Table 7: Case Study illustrating SAPO from the Musique benchmark. Key evidence is extracted from the original searched documents.

Question: Who is the president of the newly declared independent country that is part of the Commission of Truth and Friendship with the country that Tony Gunawan is from?
Ground Truth: ["Francisco Guterres"]

Model Output: Qwen2.5-14B-Instruct-SAPO

<think> To answer this question, I need to identify the country that Tony Gunawan is from, find out if it has a Commission of Truth and Friendship with any newly declared independent country, and then find out who the president of that newly declared independent country is. </think>

<search> **Tony Gunawan country of origin** </search>

<documents> **Doc 1 (Title: "Tony Gunawan"):** Tony Gunawan (, born 9 April 1975) is a Chinese descended, **Indonesian born** badminton player... Gunawan is a former Olympic gold medalist and world champion for Indonesia...
Doc 2 (Title: "Tony Gunawan"): ...Tony played for Indonesia from 1992–2001...
Doc 3 (Title: "Tony Gunawan"): ...He was a member of the world champion 2000 Indonesia Thomas Cup team... </documents>

<think> From the documents, it is clear that Tony Gunawan is from Indonesia. Now I need to find out if Indonesia has a Commission of Truth and Friendship with any newly declared independent country. </think>

<search> **Commission of Truth and Friendship Indonesia** </search>

<documents> **Doc 1 (Title: "Indonesia–Timor Leste Commission of Truth and Friendship"):** The Indonesia–Timor Leste Commission on Truth and Friendship was a truth commission established jointly by the governments of **Indonesia and East Timor** in August 2005...
Doc 2 (Title: "Indonesia–Timor Leste Commission of Truth and Friendship"): ...investigate acts of violence that occurred around the independence referendum held in East Timor in 1999... </documents>

<think> From the documents, it is clear that Indonesia has a Commission of Truth and Friendship with East Timor. Now I need to find out if East Timor is a newly declared independent country and who its president is. </think>

<search> **East Timor president** </search>

<documents> **Doc 1 (Title: "Francisco Guterres"):** Francisco Guterres, popularly known as Lú-Olo... is an East Timorese politician who has been **President of East Timor since 2017**...
Doc 3 (Title: "José Ramos-Horta"): ...José Manuel Ramos-Horta... was the President of East Timor from 20 May 2007 to 20 May 2012... </documents>

<think> From the documents, it is clear that Francisco Guterres is the current president of East Timor. However, I need to confirm if East Timor is a newly declared independent country. </think>

<search> **East Timor independence** </search>

<documents> **Doc 3 (Title: "East Timor"):** ...East Timor became the first new sovereign state of the 21st century on 20 May 2002... </documents>

<think> From the documents, it is clear that East Timor gained independence in 2002, which makes it a newly declared independent country. Therefore, the president of the newly declared independent country that is part of the Commission of Truth and Friendship with Indonesia is Francisco Guterres. </think>

<answer> **Francisco Guterres** </answer>